Business Analysis Case Study: MFU International Student Election System

1. Project Overview

The MFU International Student Election System is a business analysis case study developed to address challenges encountered during previous student elections conducted through email and Google Forms. The existing process faced issues such as duplicate voting, limited transparency, manual result management, and low student engagement.

The proposed solution is a secure web-based election platform that enables students to vote using their university credentials while providing administrators with tools to manage candidates and monitor election results efficiently. The system aims to improve election fairness, transparency, accuracy, and overall user experience.

Stakeholders

1. International Students
2. Election Committee
3. System Administrators
4. University Management

Project Goals

- Prevent duplicate voting
- Improve election transparency
- Simplify election administration
- Increase student participation
- Provide accurate and automated election results

My Role

Business Analyst | Frontend Developer

Responsibilities:

- Requirement Gathering & Analysis
- Functional Requirement Documentation
- Use Case Development
- Process Analysis
- Risk Assessment
- Stakeholder Requirement Identification

Tools Used

• Figma
• Draw.io
• Google Docs / Microsoft Word
• Jira (Backlog & User Story Management)

2. Business Problem Analysis

Problem	Impact
Duplicate voting	Unfair results
Manual counting	Slow administration
No authentication	Security risk
Poor engagement	Low participation
No real-time result tracking	Low transparency

3. Requirement Summary

Functional Requirements

- Student Login
- View Candidate Information
- Vote for Candidate
- View Election Results
- Manage Candidate Profiles

Non-Functional Requirements

- Security
- Performance
- Reliability
- Accessibility

4. User Stories

Student User Stories

US-01: View Candidates
As a student, I want to view candidate details so that I can understand their background before voting.

US-02: View Positions
As a student, I want to see all available election positions so that I can choose which position to vote for.

US-03: Prevent Duplicate Voting
As a student, I should not be able to vote more than once for the same position so that election integrity is maintained.

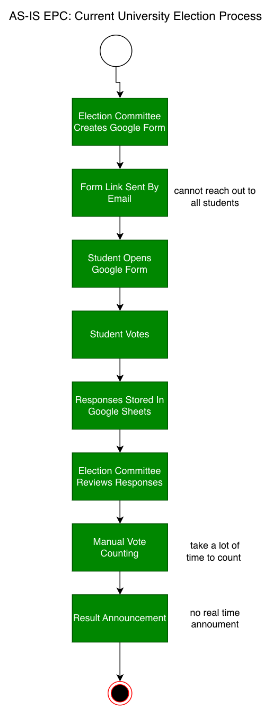
Admin User Stories

US-04: Manage Candidates
As an admin, I want to add, edit, and delete candidate information so that the election data remains accurate.

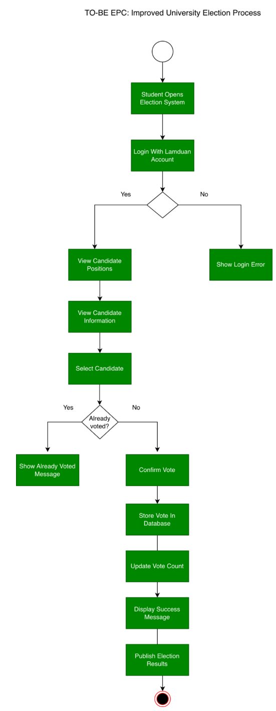
US-05: Publish Results
As an admin, I want to publish election results so that students can view the final outcome.

5. Process Flow

AS-IS Process (Current Process)

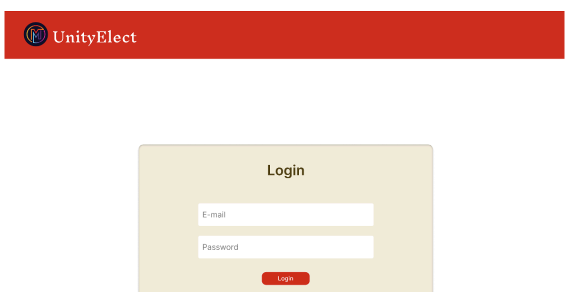


TO-BE Process (Solution)



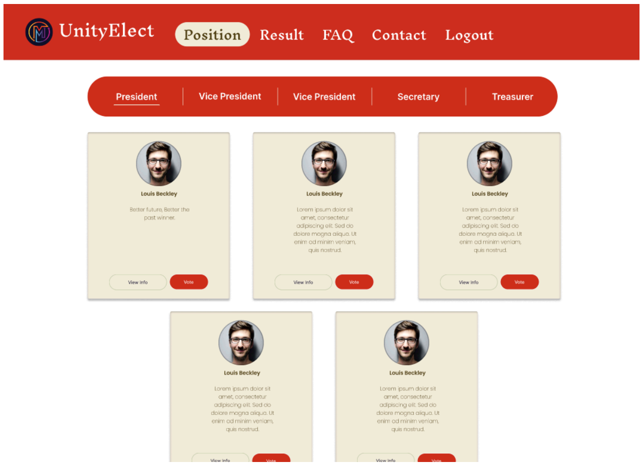
6. Wireframes/Mockups

Login Page



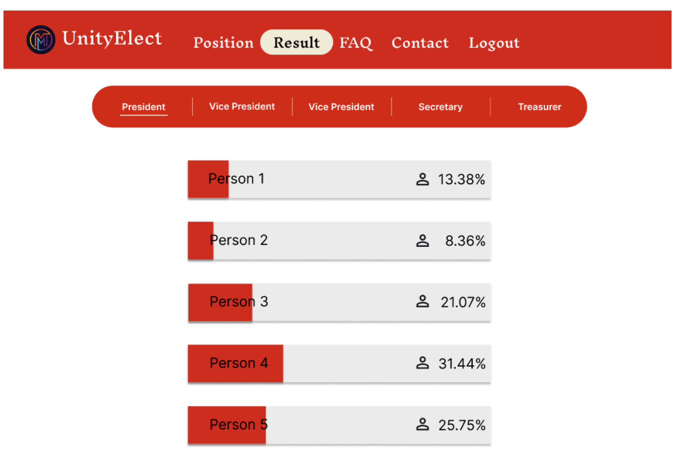
Provides secure authentication using university credentials to ensure that only eligible students and authorized administrators can access the election system.

Dashboard Page



Provides students with easy access to candidate information, election announcements, and voting functions, enabling informed participation in the election process.

Result Page



Provides real-time visibility of election outcomes through automated vote aggregation and centralized result reporting, improving transparency and decision-making.

7. Risks & Recommendations

Risk	Recommendation
Duplicate Voting	Vote Validation
Unauthorized Access	Authentication
Data Loss	Database backup
Low Participation	Email Reminders

Expected Benefits

• Prevent duplicate voting through secure authentication and vote validation.
• Improve election transparency by providing accurate and centralized election information.
• Reduce manual administrative effort in candidate and election management.
• Increase student participation through an accessible and user-friendly voting platform.
• Improve accuracy and efficiency of vote counting and result generation.
• Enhance trust in the election process through secure and auditable voting records.

8. Outcome & Lessons Learned

Through the development of the MFU International Student Election System, several important lessons were learned regarding business analysis and system design.

One key insight was the importance of clear and consistent requirements. During the analysis phase, inconsistencies were identified between business objectives and system functionalities, highlighting the need for proper requirement validation and stakeholder alignment.

Another important lesson was the value of process modeling. Comparing the AS-IS and TO-BE processes made it easier to understand inefficiencies in the existing system and demonstrate how a digital solution can improve fairness, speed, and transparency.

The project also reinforced the importance of user-centered design. Understanding the needs of students and administrators helped in designing meaningful user stories and system requirements.

Finally, this project demonstrated how structured documentation, including user stories, use cases, and diagrams, plays a critical role in communicating ideas effectively between technical and non-technical stakeholders